Summary of this week

- Last week, we studied the preliminary topics of particle filters.
- The equations that define the operations of particle filters require computing approximations to integrals.
- The brute-force methods we used last week suffer from the "curse of dimensionality."
- This week you learned how to use efficient Monte–Carlo methods to approximate integrals instead, which overcome this "curse."
- These require computing pseudo-random numbers drawn from a pdf;
  - When this isn't possible, you learned how to use pseudo-random numbers drawn from an importance density instead.
- You learned how to skip computing a difficult integral using weight normalization.
- You also learned how to represent pdfs efficiently using Dirac delta impulse functions, and how to visualize a pdf that is represented using impulse functions.

Where to from here?

- This concludes all the background material needed.
- Next week, we put all the pieces together to develop the particle filter algorithm.
- You will learn how to implement it in Octave, and apply to an example problem.
- Our first implementation will fail; we will need to investigate to discover why this is the case.
- This will lead to a small modification that makes the particle filter work robustly.
- You will also learn how to implement this modified method in Octave code; when applied to the same example, it works well.

Credits

Credits for photos in this lesson

- "Climbing upward" on slide 2: Pixabay license (`https://pixabay.com/en/service/license/`), from `https://pixabay.com/en/career-head-rise-profession-chance-1019755/`